

Low-overhead ML methods for beamforming/combining in RIS assisted millimeter-wave massive MIMO systems

Oumaïma Bounhar

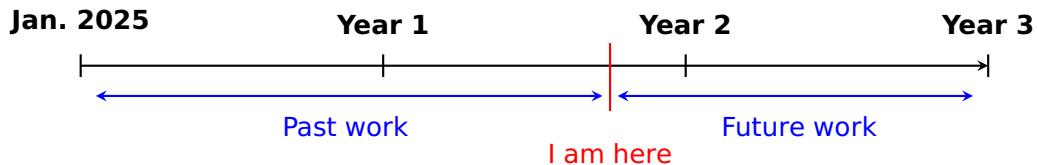
Supervised by Pr. **Ghaya Rekaya-Ben Othman**

Télécom Paris, Institut-Mines-Télécom, Palaiseau, France



21 septembre 2026

Thesis Advancement & Context





Outline

Context

System Model

Motivating Analogy

State-of-the-art

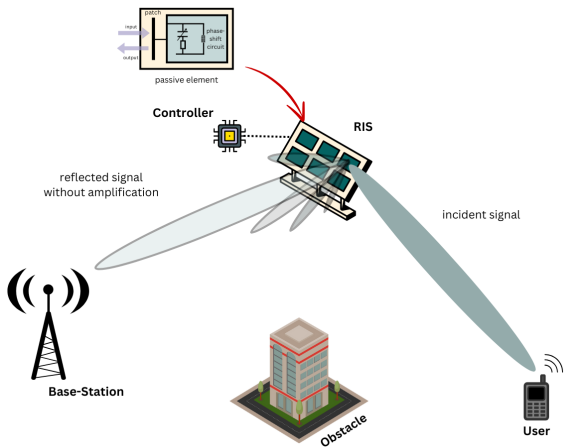
Proposed approach : Reinforcement Learning

Simulation results

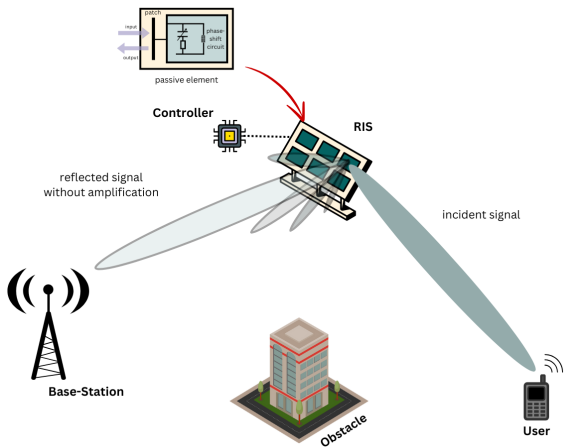
Sionna - Realistic channel generation and validation

Future Perspectives

Context

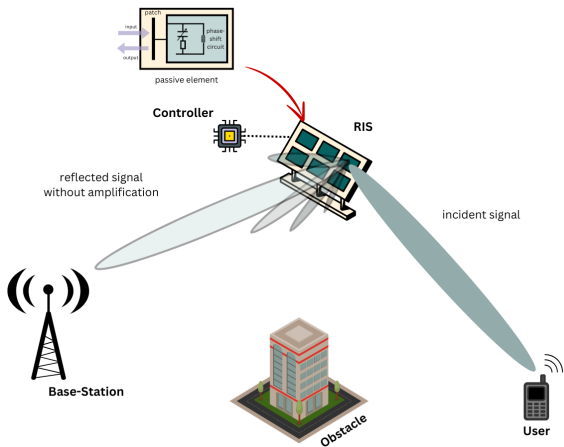


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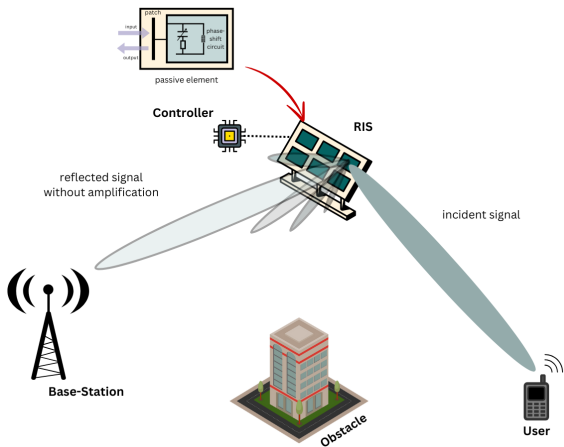
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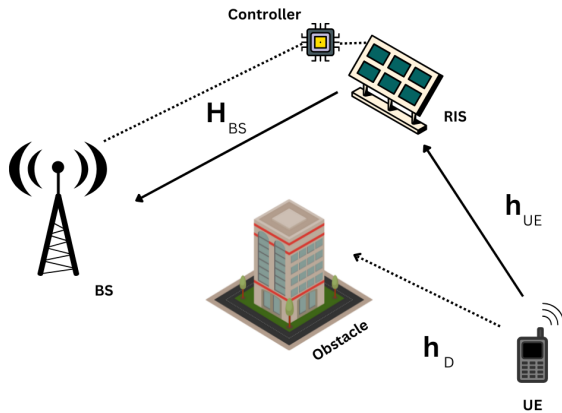
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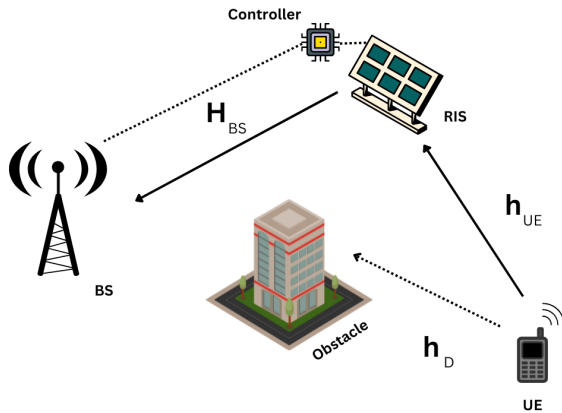
- mmWaves suffer from fading environments ;
- 5G & 6G rely on millimeter waves (mmWaves) for higher coverage ;
- RIS is a promising solution with low power consumption.

System Model



■ $N_{UE} = 1;$

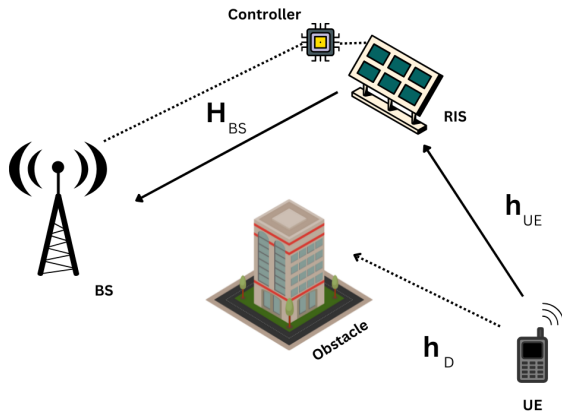
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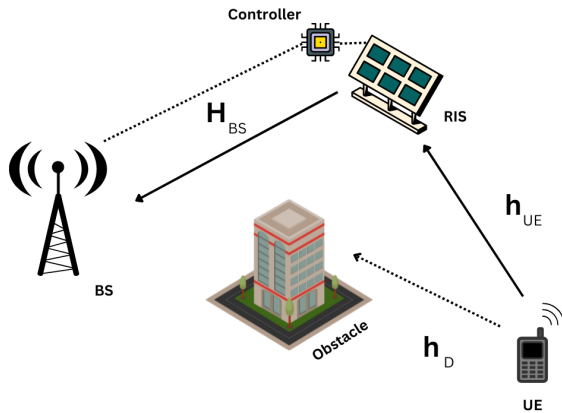


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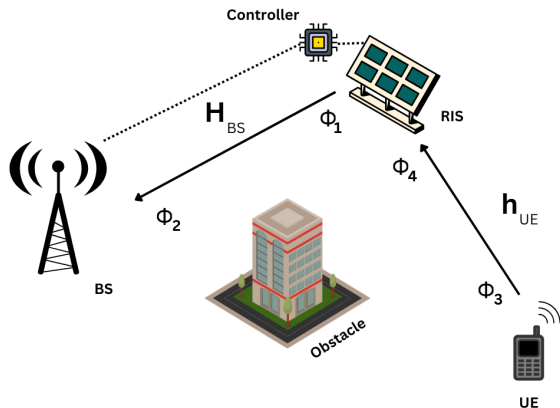
■ $N_{RIS} = 100;$

System Model



- $N_{UE} = 1$;
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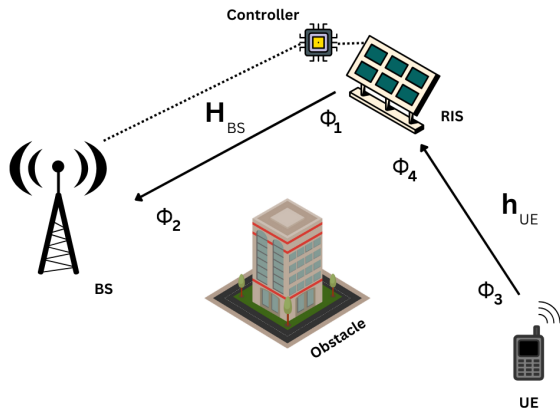
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We consider half-wave spaced ULAs :

$$\mathbf{H}_{BS} = \mathbf{a}_N(\Phi_1)\mathbf{a}_M^H(\Phi_2)$$

$$\mathbf{h}_{UE} = \mathbf{a}_1(\Phi_3)\mathbf{a}_N^H(\Phi_4)$$

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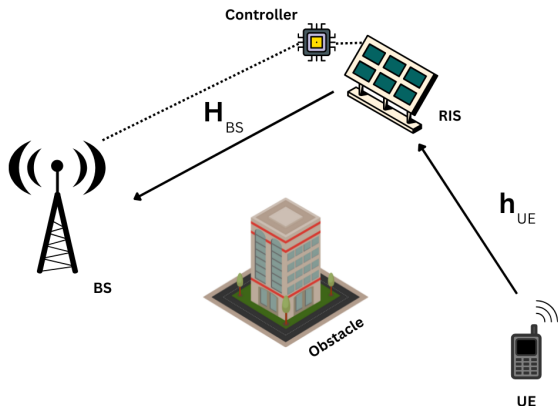
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where the steering vector is :

$$\mathbf{a}_N(\Phi) = [1, e^{j\pi \sin(\Phi)}, \dots, e^{j\pi(N-1) \sin(\Phi)}]^T$$

System Model

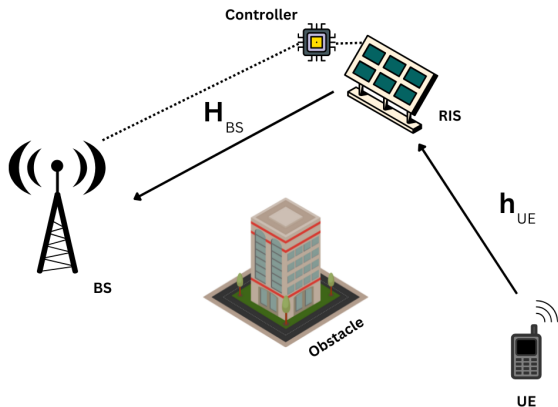


The received signal at the base station (BS) is :

$$\mathbf{y} = \sqrt{P}\mathbf{H}_{\text{BS}}\mathbf{\Theta}\mathbf{h}_{\text{UE}}s + \mathbf{w}$$

where P is the transmit power of the user, $\|s\|^2 = 1$, $\mathbf{w} \sim \mathcal{CN}(0, \sigma_w^2)$ and $\mathbf{\Theta} = \text{diag}(\mathbf{\Phi})$, with $\mathbf{\Phi} = [e^{j\theta_0}, \dots, e^{j\theta_{N_{\text{RIS}}-1}}]$, where the θ_i are the phase shifts introduced by the RIS, $i \in [0, N_{\text{RIS}} - 1]$.

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We write the full channel :

$$\mathbf{h}(\mathbf{\Phi}) = \mathbf{H}_{\text{BS}}\text{diag}(\mathbf{\Phi})\mathbf{h}_{\text{UE}}$$



System Model

Codebook design

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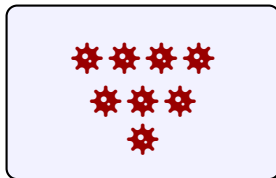
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What is the configuration of the RIS that maximises the rate ?

A Motivating Analogy

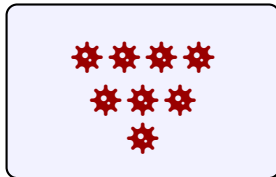
Reference Book (Dataset)



8 known diseases

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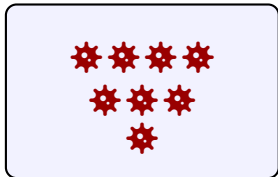
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Clinician

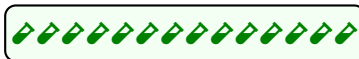
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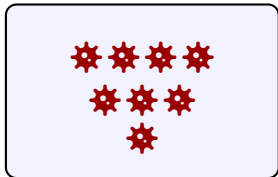
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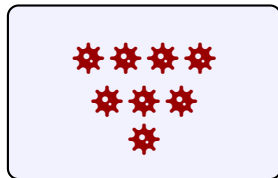
sequentially run, one at a time



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compare results

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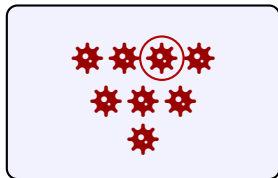
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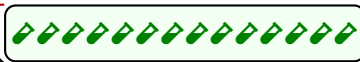


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Diagnosis found after
only a few tests!

compare results

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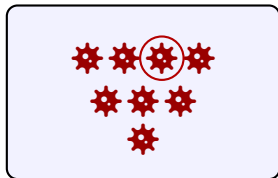
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Goal : find the right diagnosis using as few cheap tests as possible.



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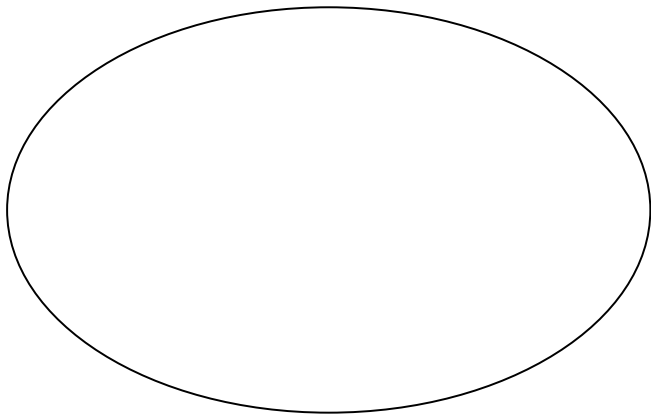
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Goal : minimize the number of pilots L_p (pilot overhead) while reliably finding Φ_{opt} .

Building the Dataset

How is a sample constructed ?

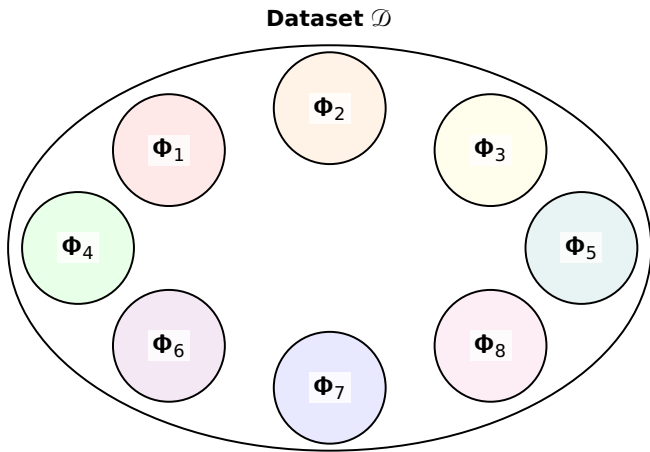
Dataset $\mathcal{D}$



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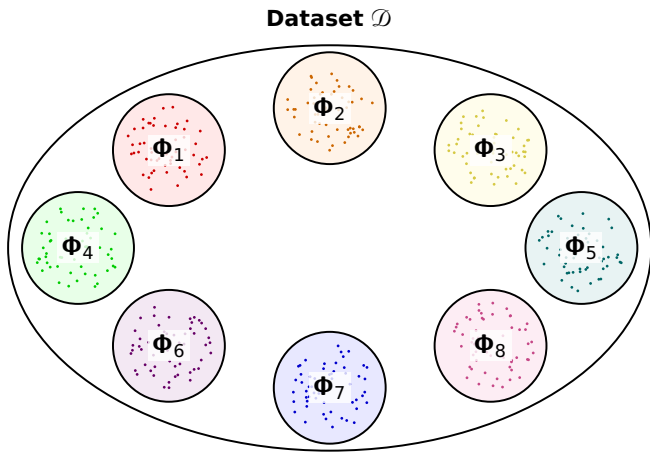
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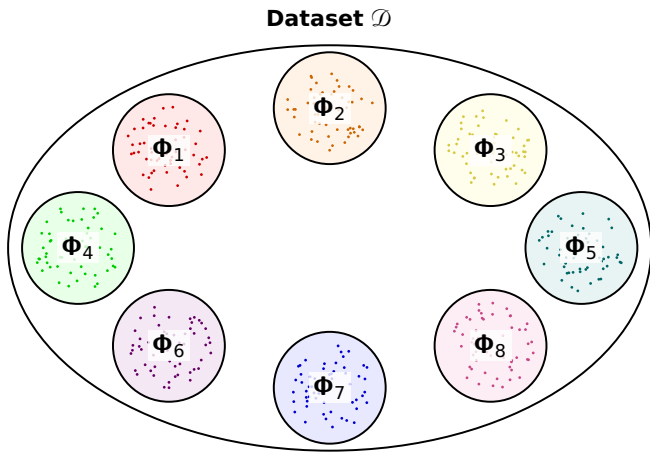
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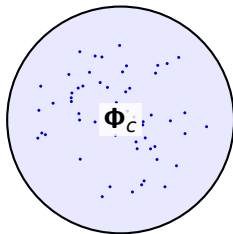


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Each class $\Phi_i \geq 200$ such samples.

The Likelihood Model

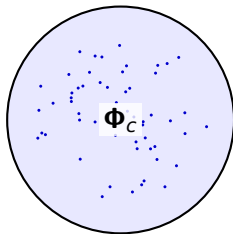
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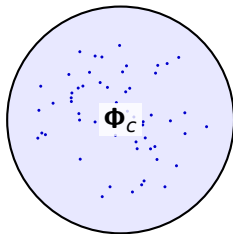


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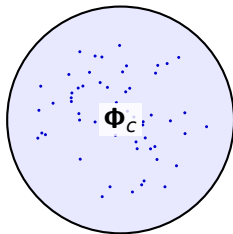
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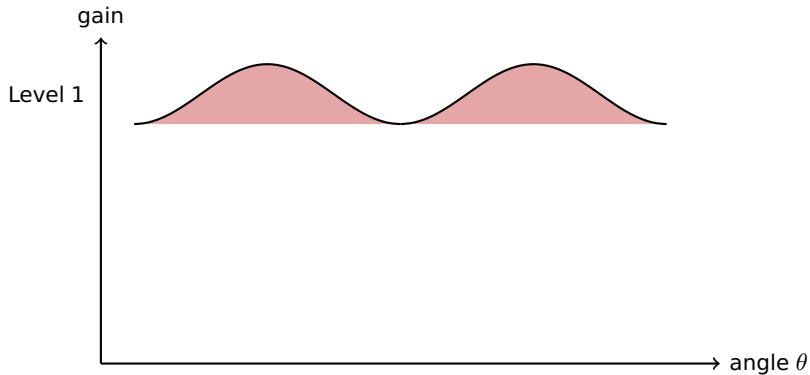


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where an observation is $(Y_k, \Psi_k) = (\text{RSE feedback, pilot codeword tested})$.

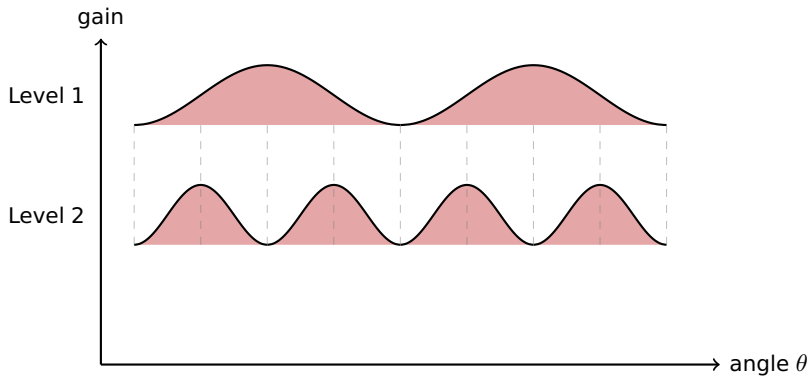
What Does Each Class Represent?

Hierarchical codebook : recursive splitting down to $N_c = 8$ classes



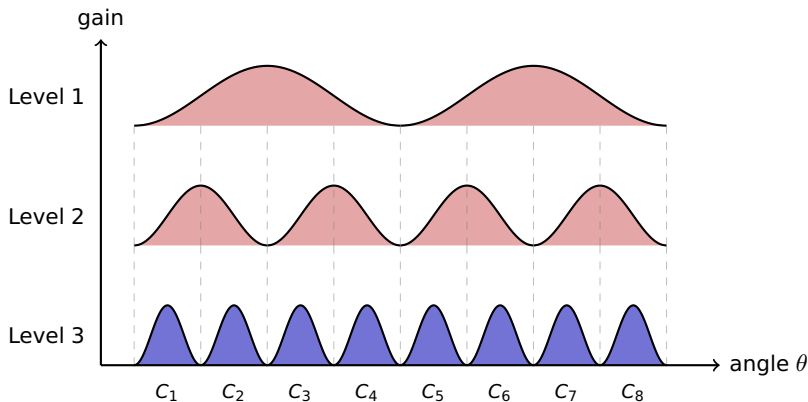
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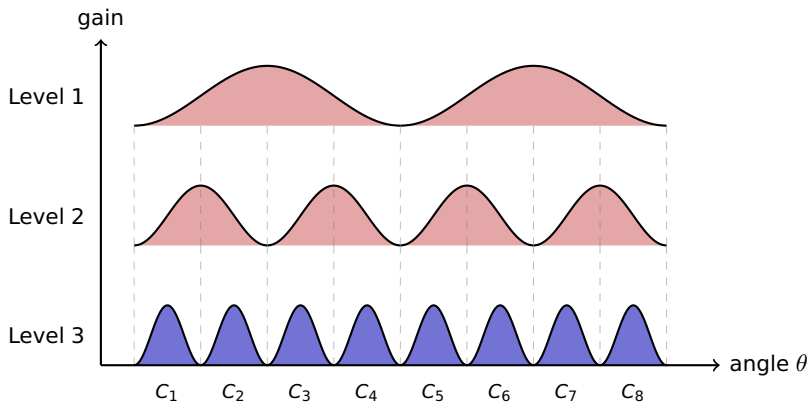
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Red : successive pilot search levels, each splitting the angular range in two.

Blue : final $N_c = 8$ narrow DFT communication beams Φ_i .

Definition (Rate)

We want to get a higher coverage, which leads to maximizing the rate :

$$R = \max_{i \in \llbracket N_c \rrbracket} \left[\log_2 \left(1 + \frac{P \|\mathbf{H}_{BS} \text{diag}(\boldsymbol{\Phi}_i) \mathbf{h}_{UE}\|_2^2}{\sigma_w^2} \right) \right]$$

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No CSI

"Blind" method.



State-of-the-art Blind Beamforming

Search for the optimal configuration (codeword $\Phi_{\text{opt}} \in \Phi_1^{N_c}$) of RIS over the possible configurations (communication codebook $\Phi_1^{N_c}$) :

- Exhaustive search over the codebook.
 - Huge overhead



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- Exhaustive search over the codebook.
 - Huge overhead
- Hierarchical : Need to create a specific codebook.
 - Low overhead.
 - Might not reach the optimal configuration because of its shape (binary tree).

State-of-the-art

Bayesian Classifier : Protocol

For a channel realization $\mathbf{H}$, we define $\Phi(\mathbf{H})$ as the class that assigns for this channel the best communication codeword $\Phi \in \Phi_1^{N_c}$:

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At each time step t , update the posteriori probability with knowledge over all received feedbacks :

$$\mathbf{p}(\Phi(\mathbf{H})|Y_1^t, \Psi_1^t) \triangleq [\mathbf{p}(\Phi_1|Y_1^t, \Psi_1^t), \dots, \mathbf{p}(\Phi_{N_c}|Y_1^t, \Psi_1^t)]$$

State-of-the-art

Naive Bayes Classifier

Classification

Let L_p be the amount of the received samples, the Bayes optimal classifier is :

$$\operatorname{argmax}_{c \in \llbracket 1, N_c \rrbracket} \mathbf{p}(\Phi(\mathbf{H}) | Y_1^{L_p}, \Psi_1^{L_p})$$

State-of-the-art

Naive Bayes Classifier

Classification

Let L_p be the amount of the received samples, the Bayes optimal classifier is :

$$\operatorname{argmax}_{c \in \llbracket 1, N_c \rrbracket} \mathbf{p}(\Phi(\mathbf{H}) | Y_1^{L_p}, \Psi_1^{L_p})$$

Class c_i is the class assigned to a UE direction for which communication codeword Φ_i maximizes the received signal strength $\|\mathbf{h}(\Phi_i)\|_2^2$.

State-of-the-art

Naive Bayes Classifier

Objective

Our main purpose is to find the optimal configuration Φ_{opt} while minimizing the number of required pilots $\Psi \in \Psi_1^{N_p}$:

$$L_{\min} = \min\{L_p : \|\mathbf{p}(\Phi(\mathbf{H})|Y_1^{L_p}, \Psi_1^{L_p})\|_{\infty} = 1\}$$

State-of-the-art

Naive Bayes Classifier

Objective

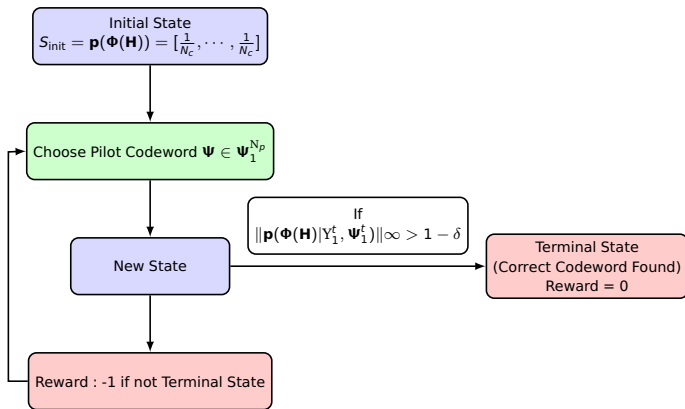
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$$L_{\min} = \min\{L_p : \|\mathbf{p}(\Phi(\mathbf{H})|Y_1^{L_p}, \Psi_1^{L_p})\|_{\infty} = 1\}$$

We introduce a fixed degree of precision $\delta \in [0; 1]$ and we need to minimize L_p st. the probability of correct classification respects a certain precision :

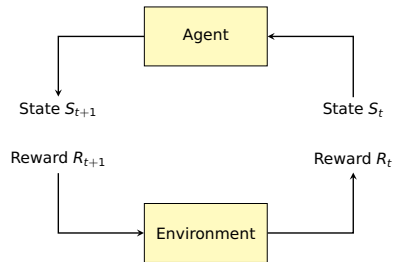
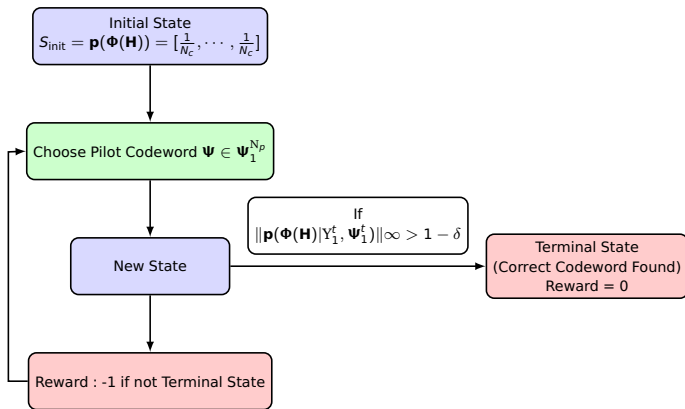
$$L_{\min} = \min\{L_p : \|\mathbf{p}(\Phi(\mathbf{H})|Y_1^{L_p}, \Psi_1^{L_p})\|_{\infty} > 1 - \delta\}$$

Proposed approach : Reinforcement Learning State, Action, and Reward Flow



Proposed approach : Reinforcement Learning

State, Action, and Reward Flow



Proposed approach : Reinforcement Learning

Stochastic shortest path problem

Policy

A policy $\pi(S)$ determines which action A is chosen in a given state S :

$$\pi(S) = A$$

Proposed approach : Reinforcement Learning

Stochastic shortest path problem

Policy

A policy $\pi(S)$ determines which action A is chosen in a given state S :

$$\pi(S) = A$$

Objective

The goal is to reach a terminal state where the probability of correct classification respects a fixed degree of precision $\delta \in [0; 1]$:

$$L_{\min} = - \sum_{t=0}^{\infty} R_t$$

Proposed approach : Reinforcement Learning

Q-Learning : Value Function and Policy

Q-function

The Q-value function represents the expected cumulative reward when taking action A_t in state S_t :

$$Q(S_t, A_t) = \mathbb{E}_{\mathbf{H}, \mathbf{w}} \left[\sum_{k=0}^{\infty} R_{t+k+1} \mid S_t = S, A_t = A \right]$$

Proposed approach : Reinforcement Learning

Q-Learning : Value Function and Policy

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Bellman Equation

The Q-values are updated using the Bellman equation :

$$Q(S_t, A_t) \leftarrow (1 - \alpha)Q(S_t, A_t) + \alpha[R_{t+1} + \max_A Q(S_{t+1}, A)]$$

where $\alpha \in [0, 1]$ is the learning rate.

Proposed approach : Reinforcement Learning

Q-Learning : Optimal policy

Optimal Policy

The optimal policy is obtained by selecting the action that maximizes the expected reward :

$$\pi^*(S) = \arg \max_A Q^*(S, A)$$

Proposed approach : Reinforcement Learning

Q-Learning : Optimal policy

Optimal Policy

The optimal policy is obtained by selecting the action that maximizes the expected reward :

$$\pi^*(S) = \arg \max_A Q^*(S, A)$$

Objective

Thus, our goal is to find the optimal policy π^* that maximizes the cumulative reward despite randomness in the channel and noise. We rewrite our objective :

$$L^* = \max_{\pi} \mathbb{E}_{\mathbf{H}, \mathbf{w}} \left[\sum_{t=0}^{\infty} R_t \right]$$



Proposed approach : Reinforcement Learning

From Q-Learning to Deep Q-Learning

- The state space was discretized [1];

[1] : T. Chêne, O. Bounhar, G. R.-B. Othman, and O. Damen, “Adaptive Passive Beamforming in RIS-Aided Communications with Q-Learning,” in 2025 IEEE Wireless Communications and Networking Conference (WCNC). Milan, Italy : IEEE, 2025, pp. 1–6.



Proposed approach : Reinforcement Learning

From Q-Learning to Deep Q-Learning

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Proposed approach : Reinforcement Learning

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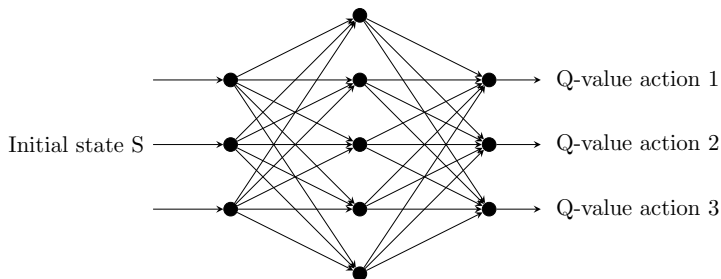
- The state space was discretized [1];
- Transitions were determined using Euclidean distances to predefined states;
- This approach became unreliable in high dimensions;
- Small variations led to misclassifications among similar codewords.

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Proposed approach : Reinforcement Learning

Deep Q-Neural Network

- Replacing the Q-table with a neural network (NN) that generalizes over continuous state spaces.





Proposed approach : Reinforcement Learning Deep Q-Neural Network

Deep Q-Learning (DQL) uses two NN :

- Evaluation network (Eval-Net) parameterized by the NN weights $\theta^{(eval)}$,



Proposed approach : Reinforcement Learning Deep Q-Neural Network

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Deep Q-Learning (DQL) uses two NN :

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- Target network (Target-Net) parameterized by NN weights $\theta^{(target)}$.

Approximation of the Q-function

The Q-function approximation can thus be expressed as :

$$Q_{eval}(S, A; \theta^{(eval)}) \approx \mathbb{E} \left[R + \gamma \max_{A'} Q_{target}(S', A'; \theta^{(target)}) \right], \quad (1)$$

Proposed approach : Reinforcement Learning

Deep Q-Learning Algorithm

Epsilon-Greedy Exploration: Agent selects action: $\arg \max Q_{eval}$ with prob. $(1 - \epsilon)$, random with prob. ϵ

Experience Replay: Store transitions (S, A, R, S') in buffer $\mathcal{M}$ and sample mini-batches

Target Network Update: Periodically copy $\theta^{eval} \rightarrow \theta^{target}$ for stable learning

Proposed approach : Reinforcement Learning

Deep Q-Neural Network

Network Optimization

The Q-network is trained by minimizing the mean squared error (MSE) loss, defined as :

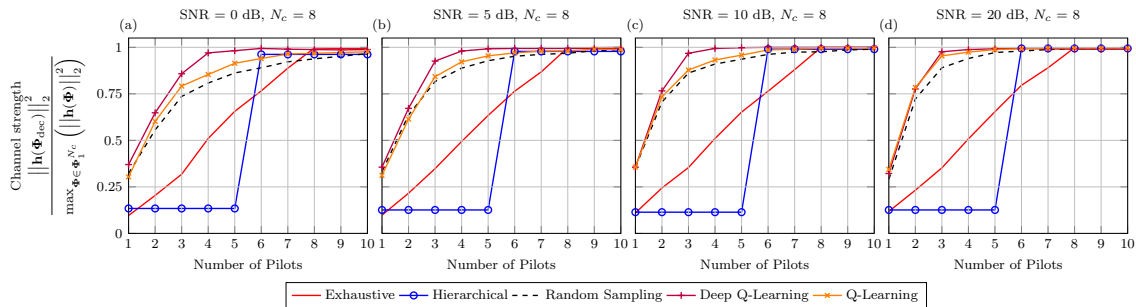
$$\mathcal{L}(\theta) = \mathbb{E}[(Q_{\text{target}} - Q_{\text{eval}}(S, A; \theta^{(\text{eval})}))^2], \quad (2)$$

with :

$$Q_{\text{target}} = R + \gamma \max_{\psi} Q_{\text{target}}(S', \psi; \theta^{(\text{target})}). \quad (3)$$

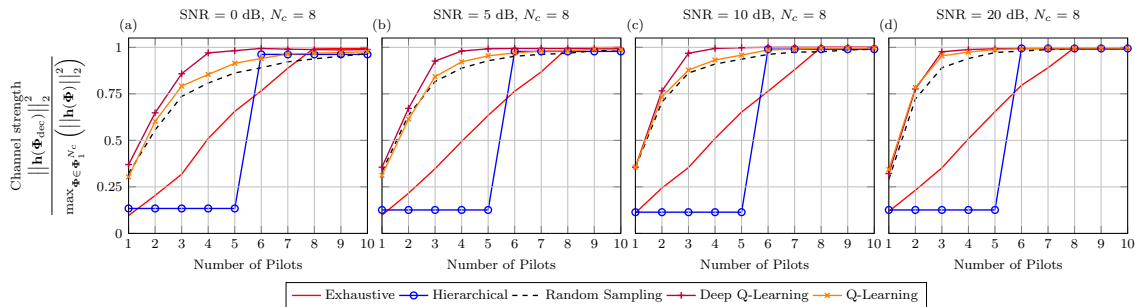
Simulation results

Comparison to other benchmarks



Simulation results

Comparison to other benchmarks

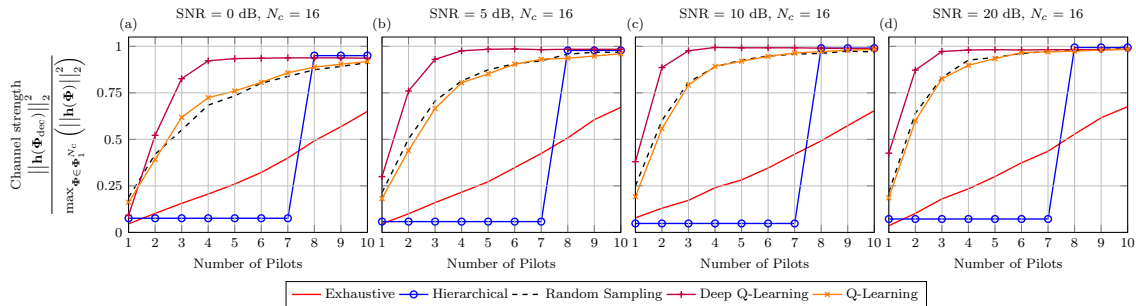


These results demonstrate that our proposed approach using Deep Q-Learning successfully minimizes pilot overhead, requiring at most $L^* = 4$ pilots in low-SNR conditions and only $L^* = 3$ pilots at moderate-to-high SNRs [2].

[2] : O. Bounhar, T. Chene, G. R.-B. Othman, and O. Damen, "Deep q-learning for passive beamforming in ris-assisted mimo systems," 2026.

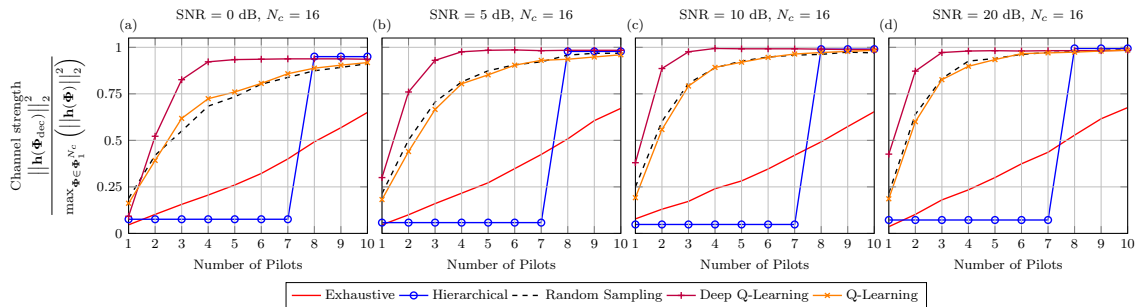
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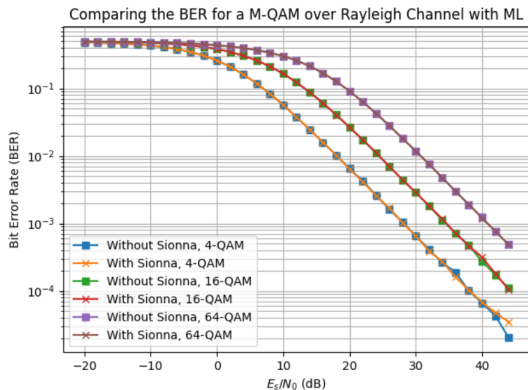
These results were extended to a larger codebook size of $N_c = 16$, where our proposed approach continues to outperform benchmarks, requiring at most $L^* = 5$ pilots in low-SNR conditions and only $L^* = 4$ pilots at moderate-to-high SNRs [2].

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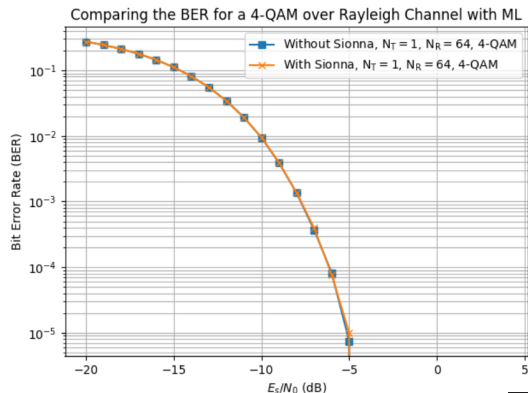
Sionna - Realistic channel generation and validation

Comparison to Rayleigh fading

1 Tx - 1 Rx



1 Tx - 64 Rx



Sionna - Realistic channel generation and validation

Primer on electromagnetics : RIS reradiated field

The reradiated field from the RIS at point S' is given by :

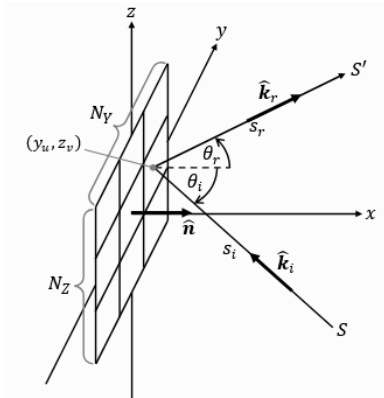
$$\mathbf{E}_r(S') = \sum_{u=1}^{N_y} \sum_{v=1}^{N_z} \Gamma_{u,v} G_{u,v} A_{u,v} \mathbf{p}_{u,v}$$

where :

$$G_{u,v} = \frac{3\lambda}{16\pi} (1 + \cos \theta_i^{u,v}) (1 + \cos \theta_r^{u,v}) ,$$

$$A_{u,v} = \frac{e^{-jk_0(s_i^{u,v} + s_r^{u,v})}}{s_i^{u,v} s_r^{u,v}} ,$$

$$\mathbf{p}_{u,v} = E_{i,\theta}(S) \hat{\theta}(\hat{\mathbf{k}}_r^{u,v}) + E_{i,\varphi}(S) \hat{\varphi}(\hat{\mathbf{k}}_r^{u,v})$$



Incident and reradiated field from a RIS

$$s_i^{u,v} = \|\mathbf{r}_{\text{RIS}}^{u,v} - \mathbf{r}_{\text{Tx}}\|_2$$

Tx-RIS distance

$$s_r^{u,v} = \|\mathbf{r}_{\text{Rx}} - \mathbf{r}_{\text{RIS}}^{u,v}\|_2$$

RIS-Rx distance

Sionna - Realistic channel generation and validation

Why geometry strongly affects the RIS channel

$$\text{Path loss} \propto \frac{1}{s_i^{u,v} s_r^{u,v}}$$

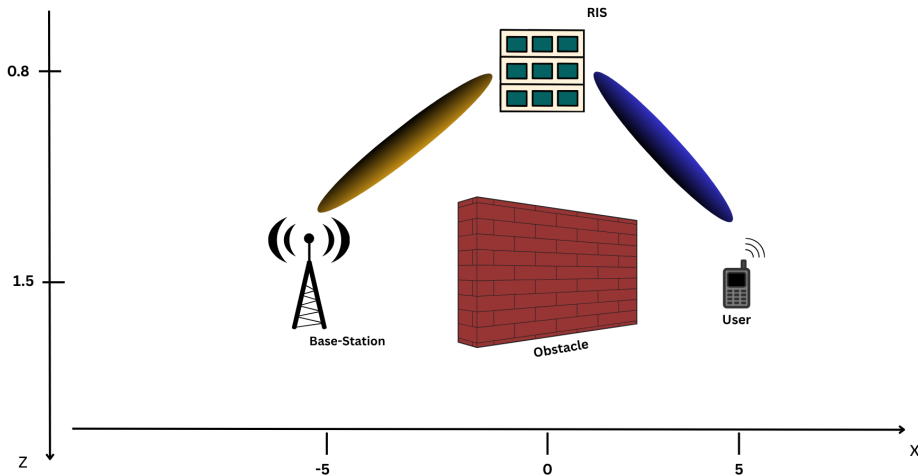
Main message

For RIS-assisted propagation, the attenuation is controlled by the both segments Tx-RIS and RIS-Rx, and it is proportional to the product of these two distances :

$$d_{\text{Tx} \rightarrow \text{RIS}} \times d_{\text{RIS} \rightarrow \text{Rx}}.$$

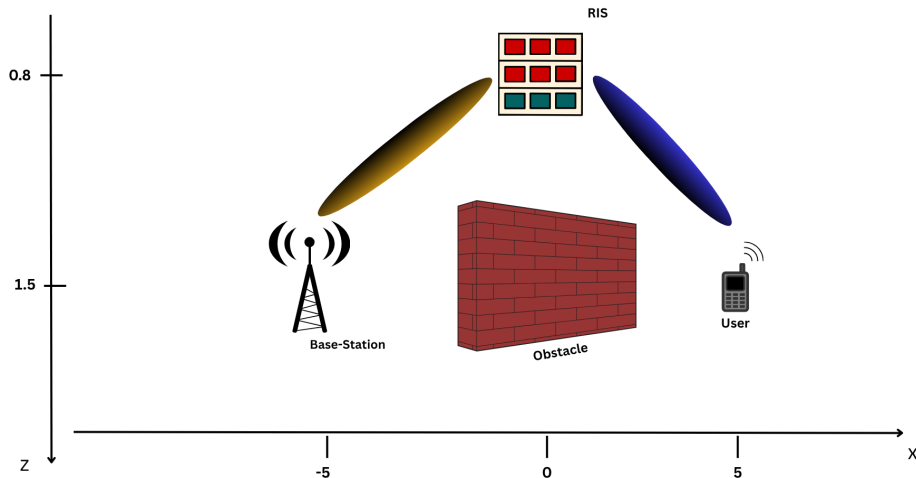
Sionna - Realistic channel generation and validation

Configuring scene geometry and RIS activation



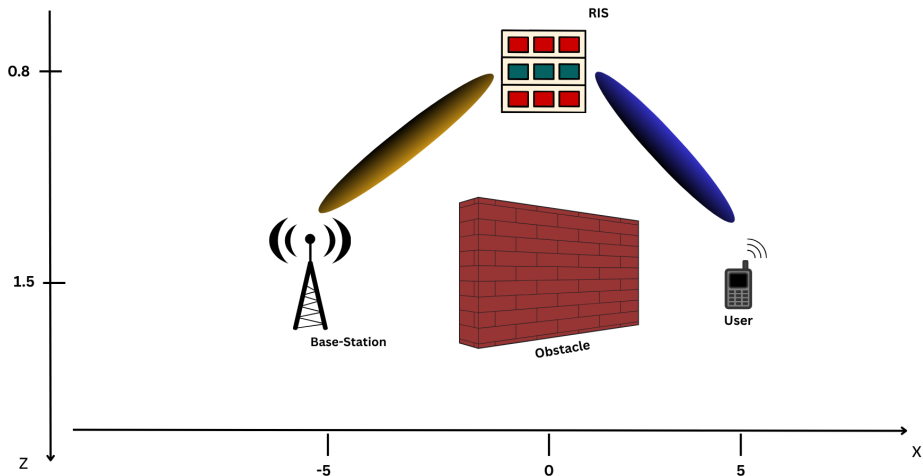
Sionna - Realistic channel generation and validation

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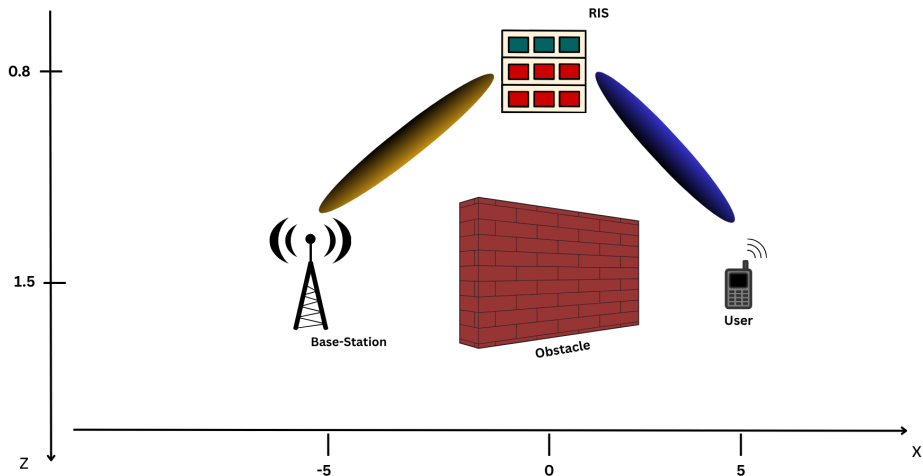
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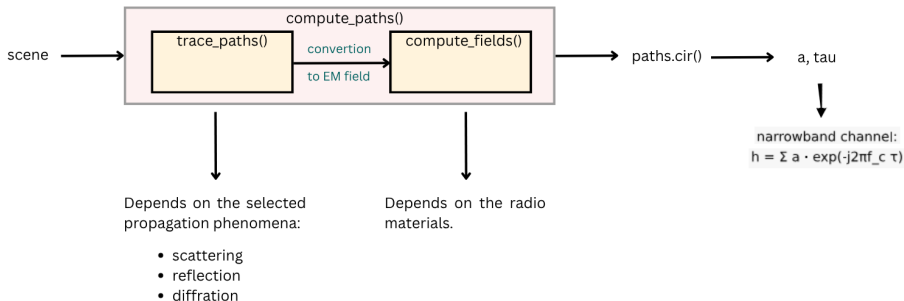
Configuring scene geometry and RIS activation



Sionna - Realistic channel generation and validation

Channel computation workflow

Computing paths goes through 2 steps:



Sionna - Realistic channel generation and validation

Toy example : 1D-DFT Codebook (3 codewords)

DFT codebook construction

For K codewords, the k -th 1D-DFT codeword is :

$$\Phi_k = [1 \quad e^{j\theta_k} \quad e^{j2\theta_k} \quad \dots \quad e^{j(N-1)\theta_k}]^T, \quad \theta_k = \frac{2\pi k}{K}.$$

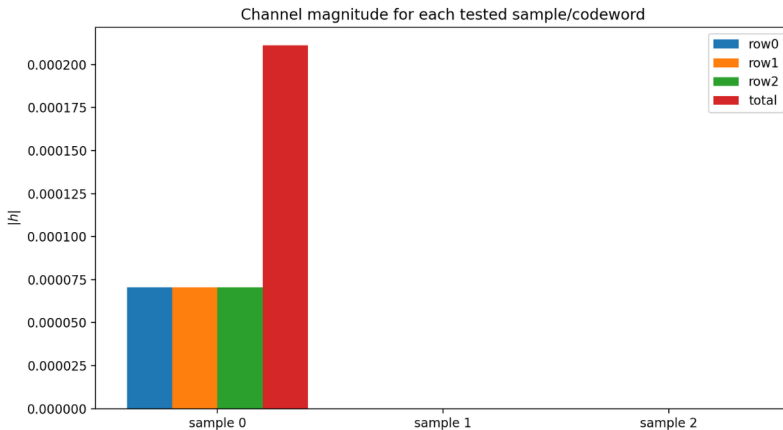
Toy example

We use a 1D-DFT codebook with $K = 3$ codewords :

$$\Phi_3^1 = [\Phi_1, \Phi_2, \Phi_3] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\frac{2\pi}{3} & \frac{2\pi}{3} \\ 0 & \frac{2\pi}{3} & -\frac{2\pi}{3} \end{bmatrix}.$$

Sionna - Realistic channel generation and validation

Toy example : 1D-DFT Codebook (3 codewords)

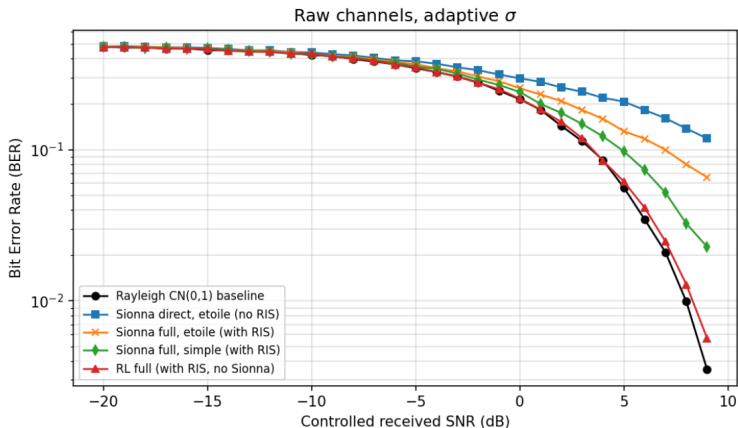


Sionna - Realistic channel generation and validation

Generated channel

4-QAM BER with ZF, $N_T = 1$, $N_R = 64$

Noise scaled as $\sigma^2 = \mathbb{E}[\|h\|^2]/\text{SNR}$



Scientific contributions



T. Chêne, O. Bounhar, G. R.-B. Othman, and O. Damen, "Adaptive Passive Beamforming in RIS-Aided Communications with Q-Learning," in **2025 IEEE Wireless Communications and Networking Conference (WCNC)**. Milan, Italy : IEEE, 2025, pp. 1–6.



O. Bounhar, T. Chêne, G. R.-B. Othman, and O. Damen, "Deep q-learning for passive beamforming in ris-assisted mimo systems," 2026.

Future Perspectives

■ *Mobile user scenario :*

- Validate the performance of our proposed approach on the generated channels with the Sionna framework.
- Handle user mobility and time-varying channels.
- Do a a real-world experiment with RIS prototypes

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■ *Multi-user extension :* Scale the study to multi-user scenarios.

Thank you ! Questions ?

Oumaïma Bounhar

Supervised by Pr. **Ghaya Rekaya-Ben Othman**

Télécom Paris, Institut-Mines-Télécom, Palaiseau, France

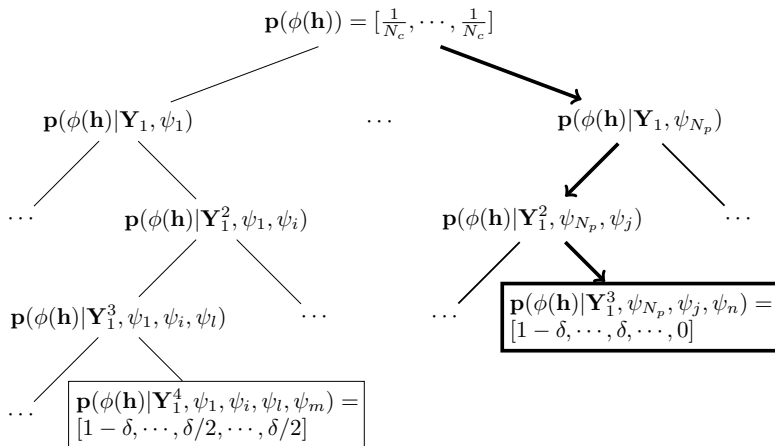


21 septembre 2026

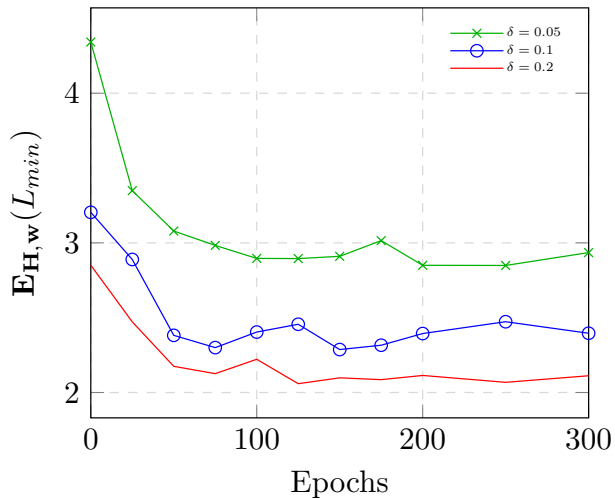
Euclidean distance

$$\mathbf{p}(\Phi(\mathbf{H}) = \Phi_k | \mathbf{Y}_1^{L_p}, \Psi_1^{L_p}) \propto \mathbf{p}(\Phi(\mathbf{H}) = \Phi_k | \mathbf{Y}_1^{L_p-1}, \Psi_1^{L_p-1}) \cdot \sum_{\mathbf{H}_1 / \mathbf{S}_T(\mathbf{H}_1)[k]=1} \frac{1}{(\mathbf{X}(\mathbf{H}_1)[L_p] - \mathbf{Y}_{L_p})^2} \quad (4)$$

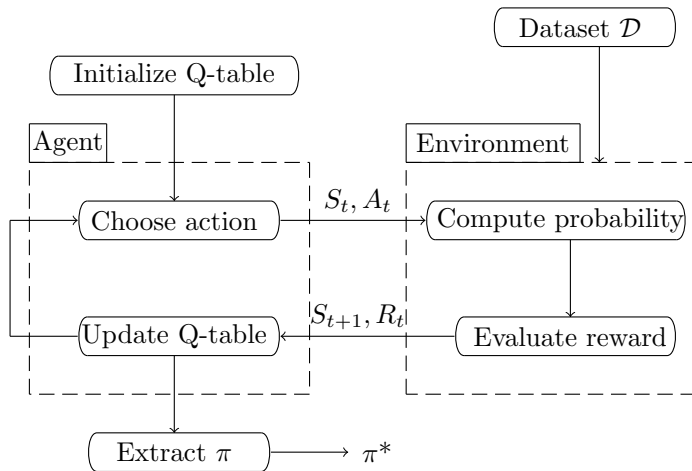
Stochastic shortest path problem



Stochastic shortest path problem

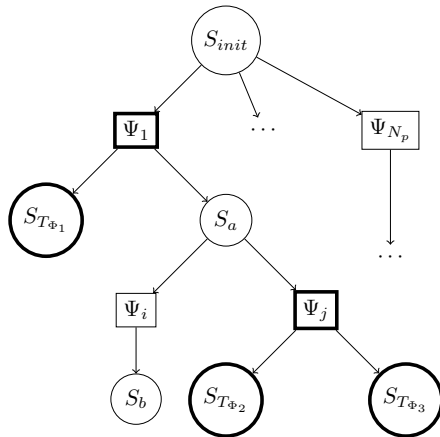


Q-Learning algorithm



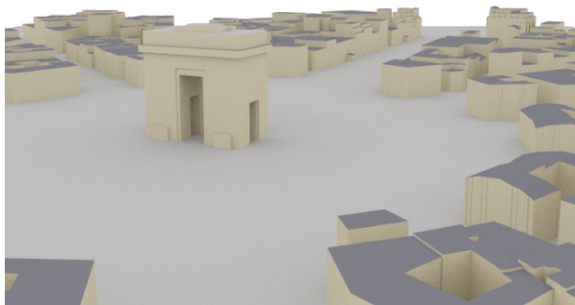
Proposed approach : Reinforcement Learning

Stochastic shortest path problem



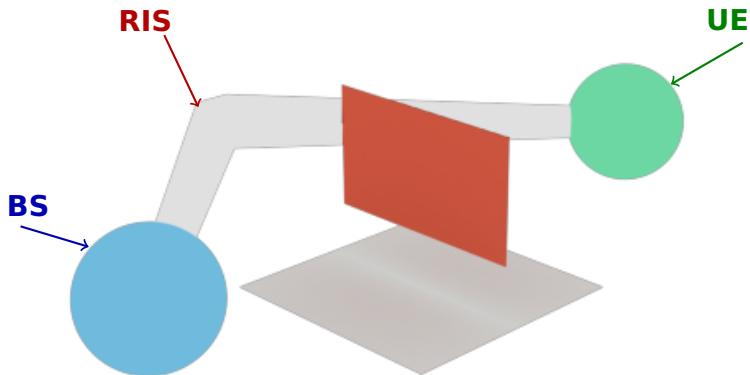
Sionna - Scene view

The area around the Arc de Triomphe in Paris



Sionna - Scene view

Simple floor-wall scenario



Visualization of the simulated scenario with a BS (64 antennas),
a RIS (1D 100 elements), and a UE (a single antenna).

$$a_i^b = a_i e^{-j2\pi f \tau_i}$$

$$H(f) = \sum_i a_i e^{-j2\pi f \tau_i}$$

$$g_{i,j} = \frac{1}{|C|} \int_{C_{i,j}} |h(s)|^2 ds$$

$$RSS_{i,j} = P_{tx} \cdot g_{i,j}$$

$$N_0 = k.T.B \text{ [W/Hz]}$$

$$\sigma^2 = 10 \log_{10}(N_0) \text{ [W]}$$

$$SNR = 10 \log_{10}\left(\frac{P_{tx} \|h\|^2}{\sigma^2}\right)$$

$$\tau_i = a_i$$

$$\sigma^2 = -174 + 10 \log_{10} B \text{ [dBm]} = -94 \text{ [dBm]}$$

$$P_{dBm} = 30 + 10 \log_{10} P_w$$

$$\text{SINR}_{i,j}^k = \frac{\text{RSS}_{i,j}^k}{N_0 + \sum_{k' \neq k} \text{RSS}_{i,j}^{k'}}$$

$$\mathbf{R} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H$$

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{W}$$

$$\mathbf{R}\mathbf{y} = \mathbf{x} + \mathbf{R}\mathbf{W}$$

$$E_{\mathbf{x}_{\text{tx}}} = \sum |\mathbf{x}_{\text{tx}}|^2$$

$$E_{\hat{\mathbf{x}}} = \sum |\hat{\mathbf{x}}|^2$$

$$\begin{aligned} E_{y_{\text{mrc}}} &= \sum |y_{\text{mrc}}|^2 \\ &= \|\mathbf{H}\|^2 E_s + \sigma_w^2 \end{aligned}$$

$$\begin{aligned} y_{\text{mrc}} &= \frac{\mathbf{H}^H \mathbf{y}}{\|\mathbf{H}\|} \\ &= \|\mathbf{H}\| \mathbf{x}_{\text{tx}} + \frac{\mathbf{H}^H \mathbf{W}}{\|\mathbf{H}\|} \end{aligned}$$

$$\frac{E_{\hat{x}}}{E_{x_{tx}}}$$

$$\|H\| E_s \ll \sigma_w^2$$

$$\|H\| E_s \gg \sigma_w^2$$

$$\sigma_w^2 = \frac{\|H\|^2 E_s}{\text{SNR}_{\text{linear}}}$$

$$\sigma_w^2 = \frac{E_s}{\text{SNR}_{\text{linear}}}$$

$$\begin{aligned} \text{SNR}_{\text{real}} &= \frac{\|H\|^2 E_s}{\sigma_w^2} \\ &= \frac{\|H\|^2 E_s}{\frac{E_s}{\text{SNR}_{\text{linear}}}} \\ &= \|H\|^2 \text{SNR}_{\text{linear}} \end{aligned}$$

$$\text{SNR}_{\text{real}} = \frac{P_{\text{tx}} \|H\|^2}{\sigma_W^2}$$

$$\Rightarrow P_{\text{tx}} = \frac{\text{SNR}_{\text{real}} \sigma_W^2}{\|H\|^2}$$

$$x = 7\pi \sin(\Phi)$$

$$y_j = r_j + \gamma Q(\Phi_{j+1}, \arg \max_{a'} Q(\Phi_{j+1}, a'; \theta); \theta^-)$$

$$\delta_t = r_t + \gamma(1 - d_t)V_{t+1} - V_t$$

$$A_t = \delta_t + \gamma \lambda (1 - e_t) A_{t+1}$$

$$R_t = A_t + V_t$$

$$r_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}$$

$$L_t^{\text{CLIP}}(\theta) = \min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)$$

$$L_{\text{actor}} = -\mathbb{E}_t [\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)]$$

$$L_{\text{critic}} = \frac{1}{2} \mathbb{E}_t [(V_t - R_t)^2]$$

$$L_{\text{critic}} = \frac{1}{2} \mathbb{E}_t [(V_{\Phi}(s_t) - R_t)^2]$$

$$H(\pi_{\theta}) = -\mathbb{E}_t \left[\sum_a \pi_{\theta}(a | s_t) \log \pi_{\theta}(a | s_t) \right]$$

$$L_{\text{total}} = L_{\text{actor}} + c_v L_{\text{critic}} + c_e L_{\text{entropy}}$$

Proximal Policy Optimization : Approximate KL Divergence

True KL divergence :

$$D_{KL}(\pi_{\text{old}} \parallel \pi_{\theta}) = \mathbb{E}_t \left[\log \frac{\pi_{\text{old}}(a_t | s_t)}{\pi_{\theta}(a_t | s_t)} \right]$$

Estimators (with $r_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\text{old}}(a_t | s_t)}$, $\log r_t = \log \pi_{\theta} - \log \pi_{\text{old}}$) :

$$k_1 = -\log r_t$$

$$k_2 = \frac{1}{2} (\log r_t)^2$$

$$k_3 = (r_t - 1) - \log r_t$$

Early stopping rule :

$$\text{if } \hat{D}_{KL} = \frac{1}{N} \sum_{t=1}^N k_3^{(t)} > \delta_{\text{target}} \Rightarrow \text{stop epoch updates}$$

$$p(Y_{1:S} | c); =; \int p(Y_{1:S} | h, \varphi_{1:S}); p(h | c); dh.$$

posterior(c); $\propto$; prior(c); $\cdot$; $\mathcal{L}(c)$ (then $+ 10^{-8}$ and normalise to sum 1)